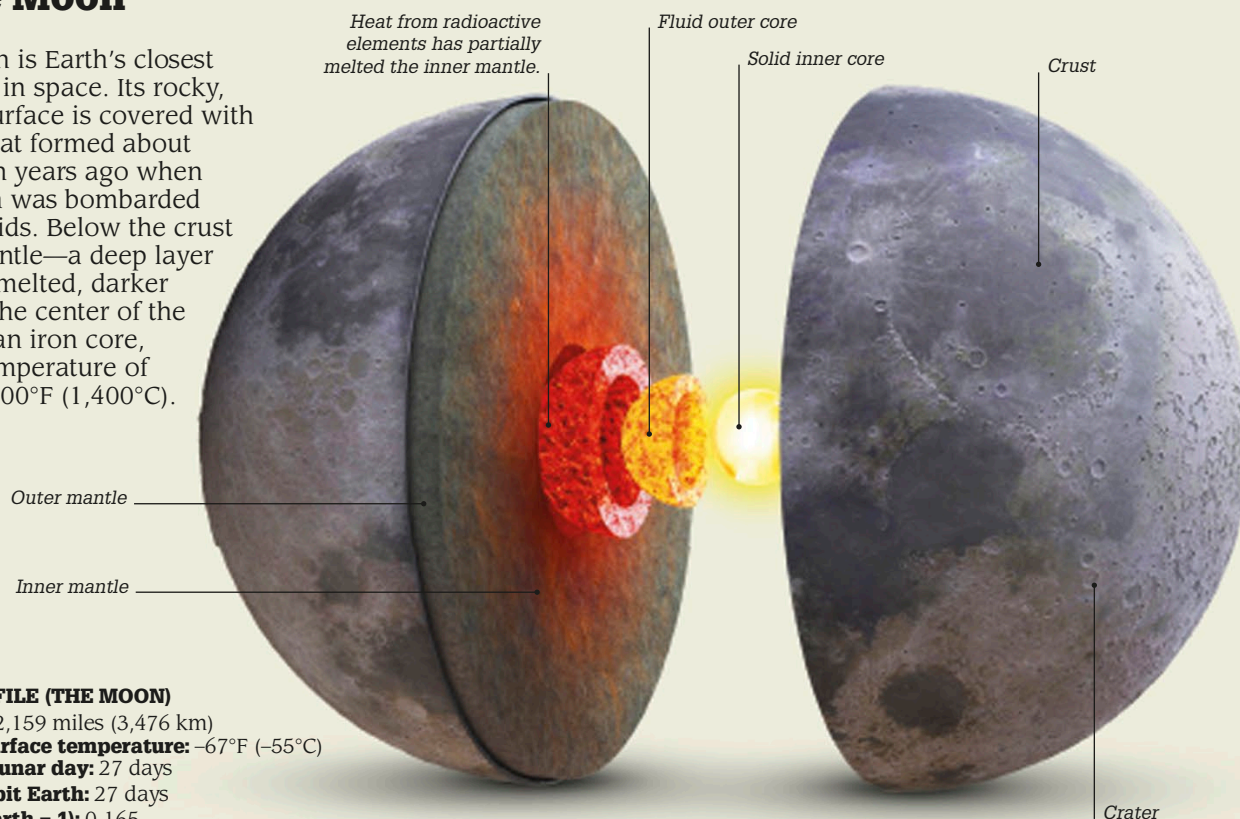


The Moon

The Moon is Earth's closest neighbor in space. Its rocky, lifeless surface is covered with craters that formed about 4.5 billion years ago when the Moon was bombarded by asteroids. Below the crust is the mantle—a deep layer of partly melted, darker rock. At the center of the Moon is an iron core, with a temperature of about 2,600°F (1,400°C).



DATA PROFILE (THE MOON)

Diameter: 2,159 miles (3,476 km)

Average surface temperature: -67°F (-55°C)

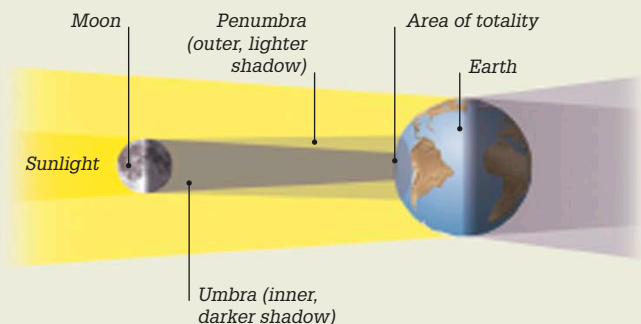
Length of lunar day: 27 days

Time to orbit Earth: 27 days

Gravity (Earth = 1): 0.165

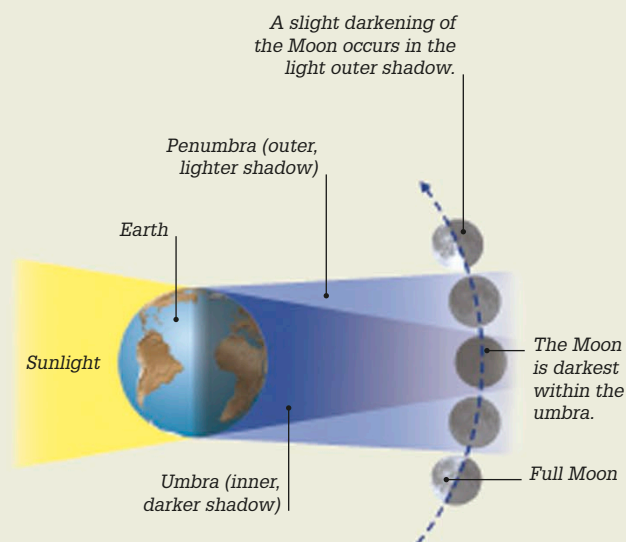
Eclipses

A solar eclipse occurs when the Moon's shadow crosses Earth's surface. A lunar eclipse takes place when the Moon moves into Earth's shadow. Up to seven solar and lunar eclipses can fall in a year.



Solar eclipse

A solar eclipse occurs when the new Moon crosses in front of the Sun and casts a shadow on Earth's surface. People in the area where the inner shadow (umbra) falls see a total eclipse. Those in the outer area (penumbra) see only a partial eclipse.



Lunar eclipse

Lunar eclipses take place only at full Moon when the Sun, Earth, and Moon are aligned. Lunar eclipses occur when the Moon moves into Earth's shadow and no sunlight reaches the Moon—so it disappears from Earth's view.